BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 070 · Lesson 070 · Heuristics and Performance Modelling

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is correlation?

Q2. What does a heuristic provide?

Q3. What is an optimisation objective?

Q4. A sorted 32-element candidate set is halved to one candidate. How many halvings are needed?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

heuristic — Estimate guiding solution search.

greedy — Choose apparently best immediate option.

performance model — Simplified resource/time estimate.

admissible — Heuristic never overestimates remaining optimal cost.

New learning

Define and predict — A heuristic is a useful estimate or rule of thumb that guides decisions when complete exploration is expensive.

Explain — A locally cheapest edge does not necessarily lead to a globally cheapest route; evaluate candidate routes as complete paths.

Evaluate — Performance modelling estimates time, memory or cost under assumptions. A predicted result requires checking against real hardware and workloads.

Worked example

Choose cheaper complete path: X→Y→G is 3+9; X→Z→G is 6+2.

1. Follow the reasoning — Calculate the entire feasible route rather than the next immediate edge.

We Do · Practise with guidance

Estimate processing 600 records at 50 records/s, then discuss what model ignores.

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600/50=12s under constant-rate assumptions; ignores setup/IO/congestion.
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Follow the reasoning

A performance model is a simplified prediction, not a guaranteed elapsed time.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Compare route A-B-G edges 2+8 against A-C-G 5+1. Calculate totals and select cheaper. [6]

Question 2

Model 1200 jobs processed at 80/s, under constant throughput: estimate duration and one assumption. [6]

Question 3

Evaluate using straight-line distance as heuristic for car routes with one-way roads and congestion. [6]

Exit Ticket · Answer before checking

A local greedy choice always produces an optimal complete path.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Statistical association, not proof of causation.

Q2. An estimate guiding an algorithm's next choice.

Q3. A quantity to minimise/maximise.

Q4. Five: 32→16→8→4→2→1.

You Do pages · Independent questions

Q1. B route10 [2], C route6 [2], choose C [1], local first-edge choice misleading [1].

Q2. 1200/80=15s [4]; stable throughput/no overhead or contention [2].

Q3. Fast estimate [1], does not model directed road restrictions [1] or congestion [1], must respect admissibility if A* optimality required [1], validate against real routes [2].

Exit Ticket · Model answer

False. Future edge costs can reverse the choice.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: